

Test

Q1. Product of Array Except Self

Given an integer array **nums**, return an array **answer** such that $\text{answer}[i]$ is equal to the product of all elements of **nums** except $\text{nums}[i]$.

You must solve it in **$O(n)$** time and without using division.

Example:

```
Input: [1, 2, 3, 4]
Output: [24, 12, 8, 6]
```

Q2. Longest Consecutive Sequence

Given an unsorted integer array **nums**, return the length of the longest consecutive elements sequence. Your solution must run in **$O(n)$** time.

Example:

```
Input: [100, 4, 200, 1, 3, 2]
Output: 4
```

Explanation: The longest consecutive sequence is [1, 2, 3, 4].

Q3. 3Sum

Given an integer array **nums**, return all unique triplets $[\text{nums}[i], \text{nums}[j], \text{nums}[k]]$ such that $i, j,$ and k are distinct and $\text{nums}[i] + \text{nums}[j] + \text{nums}[k] == 0$.

The solution must not contain duplicate triplets.

Example:

```
Input: [-1, 0, 1, 2, -1, -4]
Output: [[-1, -1, 2], [-1, 0, 1]]
```

Q4. Daily Temperatures

Given an array **temperatures**, return an array where $\text{answer}[i]$ tells you how many days you have to wait after the i -th day to get a warmer temperature.

If no future day exists, return 0.

Example:

```
Input: [73, 74, 75, 71, 69, 72, 76, 73]
Output: [1, 1, 4, 2, 1, 1, 0, 0]
```

Q5. Kth Largest Element in an Array

Given an integer array **nums** and an integer **k**, return the k -th largest element in the array.

Example:

```
Input:
nums = [3, 2, 1, 5, 6, 4]
k = 2

Output: 5
```

Q6. Merge K Sorted Linked Lists

You are given an array of **k** linked lists, where each linked list is sorted in ascending order. Merge all the linked lists into one sorted linked list.

Example:

```
Input:
[[1, 4, 5], [1, 3, 4], [2, 6]]

Output:
[1, 1, 2, 3, 4, 4, 5, 6]
```

Q7. Course Schedule II

There are **numCourses** courses labeled from 0 to numCourses - 1. prerequisites[i] = [a,b] means you must complete course b before taking course a. Return an ordering of courses you should take to finish all courses. If it is impossible, return an empty array.

Example:

```
Input:
numCourses = 4
prerequisites = [[1,0],[2,0],[3,1],[3,2]]

Output:
[0,2,1,3]
```

Q8. Lowest Common Ancestor of a Binary Tree

Given the root of a binary tree and two nodes **p** and **q**, find their lowest common ancestor.

Example:

```
Input Tree:

3
/  \
5    1
/  \ /  \
6  2 0  8
/  \
7    4

p = 5, q = 1

Output: 3
```

Q9. Word Search

Given an **m x n** grid of characters and a string **word**, return true if the word exists in the grid. The word can be constructed from sequentially adjacent cells horizontally or vertically. A cell cannot be used more than once.

Example:

```
Input:
[['A', 'B', 'C', 'E'],
 ['S', 'F', 'C', 'S'],
 ['A', 'D', 'E', 'E']]

word = "ABCCED"

Output: true
```

Q10. Longest Increasing Subsequence

Given an integer array **nums**, return the length of the longest strictly increasing subsequence.

Example:

Input:

[10, 9, 2, 5, 3, 7, 101, 18]

Output: 4

Example subsequence: [2, 3, 7, 101]